



# Business Case Study

Analysis of the Target's Data in Brazil region between the year 2016 and 2018

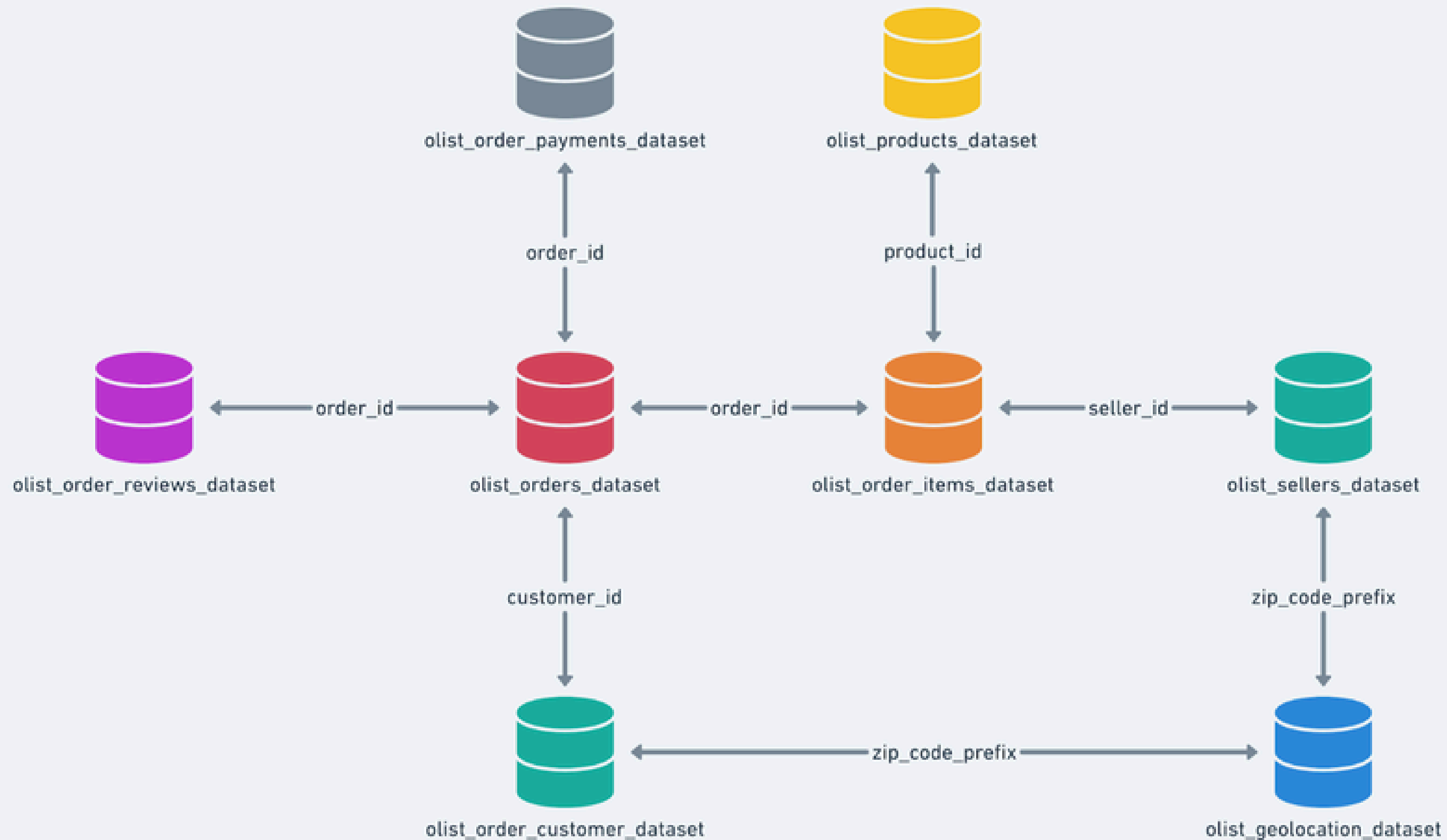
# About the case

02

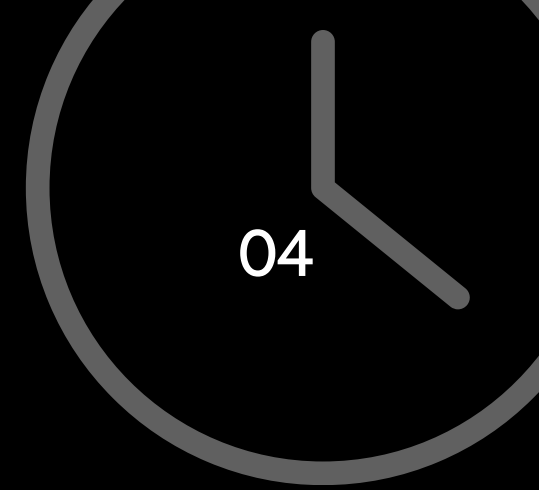
- Target is a globally renowned brand and a prominent retailer in the United States.
- This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018.
- By analysing this extensive dataset, it becomes possible to gain valuable insights into Target's operations in Brazil.



# Schema



# Q1



Data type of all columns in the "customers" table

```
SELECT
column_name,
data_type
FROM
`new-project-prathik.Target_Ecommerce.INFORMATION_SCHEMA.COLUMN`
WHERE
table_name = 'customers';
```

| Query results   |                          |           |       |      |                   |
|-----------------|--------------------------|-----------|-------|------|-------------------|
| JOB INFORMATION |                          | RESULTS   | CHART | JSON | EXECUTION DETAILS |
| Row             | column_name              | data_type |       |      |                   |
| 1               | customer_id              | STRING    |       |      |                   |
| 2               | customer_unique_id       | STRING    |       |      |                   |
| 3               | customer_zip_code_prefix | INT64     |       |      |                   |
| 4               | customer_city            | STRING    |       |      |                   |
| 5               | customer_state           | STRING    |       |      |                   |

## INSIGHTS:

- There are 5 columns in the table customers and except for customer\_zip\_code column all other column have STRING datatype and zip\_code has INT datatype.



Get the time range between which the orders were placed

```
SELECT
Count(order_id) AS No_of_orders,
MIN(order_purchase_timestamp) AS First_order_date,
MAX(order_purchase_timestamp) AS Last_order_date,
DATE_DIFF(MAX(order_purchase_timestamp) , MIN(order_purchase_timestamp) , DAY) Total_interval,
FROM
Target_Ecommerce.orders
```

| Query results   |                |                         |                         |                  |                   | SAVE RESULTS ▾  |
|-----------------|----------------|-------------------------|-------------------------|------------------|-------------------|-----------------|
| JOB INFORMATION |                | RESULTS                 | CHART                   | JSON             | EXECUTION DETAILS | EXECUTION GRAPH |
| Row             | No_of_orders ▾ | First_order_date ▾      | Last_order_date ▾       | Total_interval ▾ |                   |                 |
| 1               | 99441          | 2016-09-04 21:15:19 UTC | 2018-10-17 17:30:18 UTC | 772              |                   |                 |

- INSIGHTS:
- In the given data set the first order and the last order was placed on 4th Sep 2016 and 17th Oct 2018 respectively.
  - Total 99,441 orders were placed in a time period of 772 days.

## Count the Cities & States of customers who ordered during the given period

```
SELECT
EXTRACT(YEAR FROM order_purchase_timestamp) AS Year,
COUNT(DISTINCT customer_state) AS No_of_States,
COUNT(DISTINCT customer_city) AS No_of_Cities
FROM
Target_Ecommerce.orders o
JOIN
Target_Ecommerce.customers c ON o.customer_id = c.customer_id
GROUP BY
Year
ORDER BY
Year;
```

| JOB INFORMATION |        | RESULTS        | CHART          | JSON |
|-----------------|--------|----------------|----------------|------|
| Row             | Year ▼ | No_of_States ▼ | No_of_Cities ▼ |      |
| 1               | 2016   | 21             | 179            |      |
| 2               | 2017   | 27             | 3266           |      |
| 3               | 2018   | 27             | 3252           |      |

### INSIGHTS:

- In 2016 orders came from 21 states and 179 cities, means orders came from most of the states, but not all cities were involved.
- In 2017 and 2018 orders were made from all states and more than 55% of the cities.

### RECOMMENDATIONS:

- To reach out to remaining cities we can give advertisements in the public places.

Is there a growing trend in the no. of orders placed over the past years?

```
WITH
test1 AS
(
SELECT
EXTRACT(YEAR FROM order_purchase_timestamp) AS Year,
EXTRACT(Month FROM order_purchase_timestamp) AS Month,
COUNT(DISTINCT order_id) AS Current_Month_orders
FROM
Target_Ecommerce.orders o
LEFT JOIN
Target_Ecommerce.customers c ON o.customer_id = c.customer_id
LEFT JOIN
Target_Ecommerce.geolocation g ON c.customer_zip_code_prefix = g.geolocation_zip_code_prefix
GROUP BY
Year,Month
)
SELECT
*,
ROUND(((Current_Month_orders-Previous_Month_orders) / Previous_Month_orders) * 100, 2)
AS Percentage_Change
FROM
(
SELECT
Year,
Month,
Lead(Current_Month_orders) OVER(ORDER BY Year DESC, Month DESC) AS Previous_Month_orders,
Current_Month_orders
FROM
test1
) t
Order by
Year, Month;
```

| JOB INFORMATION |      | RESULTS |                       | CHART                | JSON              | EXECUTION DETAILS |
|-----------------|------|---------|-----------------------|----------------------|-------------------|-------------------|
| Row             | Year | Month   | Previous_Month_orders | Current_Month_orders | Percentage_Change |                   |
| 1               | 2016 | 9       | null                  | 4                    | null              |                   |
| 2               | 2016 | 10      | 4                     | 324                  | 8000.0            |                   |
| 3               | 2016 | 12      | 324                   | 1                    | -99.69            |                   |
| 4               | 2017 | 1       | 1                     | 800                  | 79900.0           |                   |
| 5               | 2017 | 2       | 800                   | 1780                 | 122.5             |                   |
| 6               | 2017 | 3       | 1780                  | 2682                 | 50.67             |                   |
| 7               | 2017 | 4       | 2682                  | 2404                 | -10.37            |                   |
| 8               | 2017 | 5       | 2404                  | 3700                 | 53.91             |                   |
| 9               | 2017 | 6       | 3700                  | 3245                 | -12.3             |                   |
| 10              | 2017 | 7       | 3245                  | 4026                 | 24.07             |                   |
| 11              | 2017 | 8       | 4026                  | 4331                 | 7.58              |                   |
| 12              | 2017 | 9       | 4331                  | 4285                 | -1.06             |                   |
| 13              | 2017 | 10      | 4285                  | 4631                 | 8.07              |                   |

- INSIGHTS:
- Starting from Sep 2016 to Apr 2018 there is a growing trend, there are dips in the number of orders in between, but the number of orders had clocked to 7,000 from being in 1,000.
  - But from Apr 2018 we can see a regular decrease in the number of orders placed, and it as come down to 16 in the month of Sep with a sudden dip(May be there is a issue with the data as we can only observe a slight ups and down in the previous months).
- RECOMMENDATIONS:
- Check the data of Sep and Oct month for the year 2018 to verify the data is correct.
  - If it is true then we need see why there is sudden drastic dip and take actions accordingly.

Can we see some kind of monthly seasonality in terms of the no. of orders being placed

```
WITH
test1 AS
(
SELECT
EXTRACT(YEAR FROM order_purchase_timestamp) AS Year,
EXTRACT(Month FROM order_purchase_timestamp) AS Month,
COUNT(DISTINCT order_id) AS No_of_orders
FROM
Target_Ecommerce.orders o
LEFT JOIN
Target_Ecommerce.customers c ON o.customer_id = c.customer_id
LEFT JOIN
Target_Ecommerce.geolocation g ON c.customer_zip_code_prefix = g.geolocation_zip_code_prefix
GROUP BY
Year,Month
)
SELECT
Year,
Month,
No_of_orders,
Round ( SUM (No_of_orders) OVER ( PARTITION BY Month ORDER BY Month DESC), 0) AS Monthly_Avg
FROM
test1
ORDER BY
Month, Year;
```

| Query results   |      |         |              |             |                   |
|-----------------|------|---------|--------------|-------------|-------------------|
| JOB INFORMATION |      | RESULTS | CHART        | JSON        | EXECUTION DETAILS |
| Row             | Year | Month   | No_of_orders | Monthly_Avg |                   |
| 1               | 2017 | 1       | 800          | 8069.0      |                   |
| 2               | 2018 | 1       | 7269         | 8069.0      |                   |
| 3               | 2017 | 2       | 1780         | 8508.0      |                   |
| 4               | 2018 | 2       | 6728         | 8508.0      |                   |
| 5               | 2017 | 3       | 2682         | 9893.0      |                   |
| 6               | 2018 | 3       | 7211         | 9893.0      |                   |
| 7               | 2017 | 4       | 2404         | 9343.0      |                   |
| 8               | 2018 | 4       | 6939         | 9343.0      |                   |
| 9               | 2017 | 5       | 3700         | 10573.0     |                   |
| 10              | 2018 | 5       | 6873         | 10573.0     |                   |
| 11              | 2017 | 6       | 3245         | 9412.0      |                   |

- INSIGHTS:
- Average number of orders in the month May, Jul, Aug are high compared to other months.
  - Number of orders placed in Nov and Dec are low.
  - Other months have good average number of orders.
- RECOMMENDATIONS:
- We can provide quarterly or yearly offers like year-end-sale to clear up the inventory stocks.

During what time of the day, do the Brazilian customers mostly place their orders?

```
WITH
test1 AS
(
SELECT
EXTRACT(HOUR FROM order_purchase_timestamp) AS Hour
FROM Target_Ecommerce.orders
)
SELECT
CASE
WHEN Hour IN (0,1,2,3,4,5) THEN 'Dawn'
WHEN Hour IN (6,7,8,9,10,11) THEN 'Morning'
WHEN Hour IN (12,13,14,15,16,17) THEN 'Afternoon'
ELSE 'Night'
END AS Interval1,
COUNT(*) AS Total_orders
FROM
test1
GROUP BY
Interval1
ORDER BY
Total_orders DESC;
```

| JOB INFORMATION |           | RESULTS      | CHART | JSOI |
|-----------------|-----------|--------------|-------|------|
| Row             | Interval1 | Total_orders |       |      |
| 1               | Afternoon | 38361        |       |      |
| 2               | Night     | 34100        |       |      |
| 3               | Morning   | 22240        |       |      |
| 4               | Dawn      | 4740         |       |      |

Insights:

- We can see most orders are placed during after afternoon and night, which means users are mostly active during this period.
- Highest number of orders came in the afternoon.

Recommendations:

- We can give calculated discounts and offers during morning to increase the sales in the that period.

# Q7

10

Find the month-on-month no. of orders placed using different payment types

```
WITH
test1 AS
(
  SELECT
    c.customer_state AS State,
    EXTRACT(Month FROM order_purchase_timestamp) AS Month,
    COUNT(DISTINCT order_id) AS No_of_orders
  FROM
    Target_Ecommerce.orders o
  JOIN
    Target_Ecommerce.customers c ON o.customer_id = c.customer_id
  GROUP BY
    State,Month
)

SELECT
  State,
  Month,
  No_of_orders,
  SUM(No_of_orders) OVER(PARTITION BY STATE) AS Total_orders_in_state
FROM
  test1
ORDER BY
  State, Month;
```

| JOB INFORMATION |       | RESULTS         | CHART        | JSON |
|-----------------|-------|-----------------|--------------|------|
| Row             | Month | Mode_Of_Payment | No_of_orders |      |
| 1               | 1     | UPI             | 1715         |      |
| 2               | 1     | credit_card     | 6093         |      |
| 3               | 1     | debit_card      | 118          |      |
| 4               | 1     | voucher         | 337          |      |
| 5               | 2     | UPI             | 1723         |      |
| 6               | 2     | credit_card     | 6582         |      |
| 7               | 2     | debit_card      | 82           |      |
| 8               | 2     | voucher         | 288          |      |
| 9               | 3     | UPI             | 1942         |      |
| 10              | 3     | credit_card     | 7682         |      |
| 11              | 3     | debit_card      | 109          |      |
| 12              | 3     | voucher         | 395          |      |
| 13              | 4     | UPI             | 1783         |      |

- INSIGHTS:
- We can the majority of the payments were made using credit cards.
  - Number of UPI payments are less than 30% for the credit card payments.

- RECOMMENDATIONS:
- Get in touch with banks and establish a tie-ups and provide EMI offers to the customers.
  - Give day wise offers on bank credit cards, so we can build a wider association with different banks.

# Q8

Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

```
WITH test1 AS
(
SELECT
  EXTRACT(YEAR FROM order_purchase_timestamp) AS Year,
  EXTRACT(Month FROM order_purchase_timestamp) AS Month,
  ROUND(SUM(payment_value), 2) AS Current_Month_Cost_of_orders
FROM
  Target_Ecommerce.orders o
JOIN
  Target_Ecommerce.payments p ON o.order_id = p.order_id

GROUP BY
  Year, Month
),
test2 AS
(SELECT
  *,
  LAG(Current_Month_Cost_of_orders, 1) OVER(ORDER BY Year, Month ) AS Previous_Month_Cost_of_orders
FROM
  test1
)
SELECT
  *,
  ROUND((((Current_Month_Cost_of_orders - Previous_Month_Cost_of_orders) / Previous_Month_Cost_of_orders) * 100 , 2) AS Monthly_Percentage_Change
FROM
  test2
WHERE
  Year IN (2017,2018)
AND
  Month Between 1 and 8
ORDER BY
  Year DESC, Month DESC;
```

Insights:

- In Feb 2017 we can see the 110% of increase in the cost of orders, apart from this we can see Mar and May in 2017 have 54% and 41% increase respectively.
- During the year 2017 we can see all most all month having at least 10% increase in the cost of orders.
  - In the year 2018 we can observe the dips in the % of cost of orders.

Recommendations:

- We can try giving combo offers to increase the cost of orders.

| JOB INFORMATION |      |       | RESULTS                      | CHART                         | JSON                      | EXECUTION DETAILS | EXECUTION GRAPH |
|-----------------|------|-------|------------------------------|-------------------------------|---------------------------|-------------------|-----------------|
| Row             | Year | Month | Current_Month_Cost_of_orders | Previous_Month_Cost_of_orders | Monthly_Percentage_Change |                   |                 |
| 1               | 2018 | 8     | 1022425.32                   | 1066540.75                    | -4.14                     |                   |                 |
| 2               | 2018 | 7     | 1066540.75                   | 1023880.5                     | 4.17                      |                   |                 |
| 3               | 2018 | 6     | 1023880.5                    | 1153982.15                    | -11.27                    |                   |                 |
| 4               | 2018 | 5     | 1153982.15                   | 1160785.48                    | -0.59                     |                   |                 |
| 5               | 2018 | 4     | 1160785.48                   | 1159652.12                    | 0.1                       |                   |                 |
| 6               | 2018 | 3     | 1159652.12                   | 992463.34                     | 16.85                     |                   |                 |
| 7               | 2018 | 2     | 992463.34                    | 1115004.18                    | -10.99                    |                   |                 |
| 8               | 2018 | 1     | 1115004.18                   | 878401.48                     | 26.94                     |                   |                 |
| 9               | 2017 | 8     | 674396.32                    | 592382.92                     | 13.84                     |                   |                 |
| 10              | 2017 | 7     | 592382.92                    | 511276.38                     | 15.86                     |                   |                 |
| 11              | 2017 | 6     | 511276.38                    | 592918.82                     | -13.77                    |                   |                 |
| 12              | 2017 | 5     | 592918.82                    | 417788.03                     | 41.92                     |                   |                 |
| 13              | 2017 | 4     | 417788.03                    | 449863.6                      | -7.13                     |                   |                 |

# Q9

12

Calculate the Total & Average value of order price & value of order freight for each state.

```
SELECT
customer_state AS State,
ROUND(SUM(freight_value), 2) AS Total_freight_price,
ROUND(AVG(freight_value) , 2) AS Avg_freight_price,
ROUND(SUM(price), 2) AS Total_order_price,
ROUND(AVG(price) , 2) AS Avg_order_price
FROM
Target_Ecommerce.order_items o1
JOIN
Target_Ecommerce.orders o2 ON o1.order_id = o2.order_id
JOIN
Target_Ecommerce.customers c ON o2.customer_id = c.customer_id
GROUP BY
State
ORDER BY
State;
```

| JOB INFORMATION |       | RESULTS             | CHART             | JSON              | EXECUTION DETAILS | EXECUTION GRAPH |
|-----------------|-------|---------------------|-------------------|-------------------|-------------------|-----------------|
| Row             | State | Total_freight_price | Avg_freight_price | Total_order_price | Avg_order_price   |                 |
| 1               | AC    | 3686.75             | 40.07             | 15982.95          | 173.73            |                 |
| 2               | AL    | 15914.59            | 35.84             | 80314.81          | 180.89            |                 |
| 3               | AM    | 5478.89             | 33.21             | 22356.84          | 135.5             |                 |
| 4               | AP    | 2788.5              | 34.01             | 13474.3           | 164.32            |                 |
| 5               | BA    | 100156.68           | 26.36             | 511349.99         | 134.6             |                 |
| 6               | CE    | 48351.59            | 32.71             | 227254.71         | 153.76            |                 |
| 7               | DF    | 50625.5             | 21.04             | 302603.94         | 125.77            |                 |
| 8               | ES    | 49764.6             | 22.06             | 275037.31         | 121.91            |                 |
| 9               | GO    | 53114.98            | 22.77             | 294591.95         | 126.27            |                 |
| 10              | MA    | 31523.77            | 38.26             | 119648.22         | 145.2             |                 |

Insights:

- We can see with 191.48 PB has the highest average order price, having 140 to 150 has the average order price throughout Brazil.
  - There are few states like AP and RR where they have higher average order price even though having lesser total order price, which means the price per product is higher in these states.
- As the number of orders is higher in SP, RJ, MG, RS, and PR the total freight price is also higher

Recommendations:

- When a customer adds a product to cart we can provide associated recommendations to increase the order price.
- Combo offers like Buy 2 get 1 etc. can be tried.

# Q10

Find out the top 5 states with the highest & lowest average freight value.

| Row | SL_NO | State | Highest_Avg_freight_price | SL_NO_1 | State_1 | Lowest_Avg_freight_price |
|-----|-------|-------|---------------------------|---------|---------|--------------------------|
| 1   | 1     | RR    | 42.98                     | 1       | SP      | 15.15                    |
| 2   | 2     | PB    | 42.72                     | 2       | PR      | 20.53                    |
| 3   | 3     | RO    | 41.07                     | 3       | MG      | 20.63                    |
| 4   | 4     | AC    | 40.07                     | 4       | RJ      | 20.96                    |
| 5   | 5     | PI    | 39.15                     | 5       | DF      | 21.04                    |

Insights:

- The average freight price varies between the range of 42 to 15.

Recommendations:

- We can reducing the freight price in the state having high price by having a decentralised or local inventory.
- Sourcing and supplying locally.
- By getting tied by with delivery partners.

```
WITH
Highest AS
(
SELECT
ROW_NUMBER() OVER(ORDER BY ROUND(AVG(freight_value) , 2) DESC ) AS SL_NO
customer_state AS State,
ROUND(AVG(freight_value) , 2) AS Avg_freight_price
FROM
Target_Ecommerce.order_items o1
JOIN
Target_Ecommerce.orders o2 ON o1.order_id = o2.order_id
JOIN
Target_Ecommerce.customers c ON o2.customer_id = c.customer_id
GROUP BY
State
ORDER BY
Avg_freight_price DESC
LIMIT 5
),

Lowest AS
(
SELECT
ROW_NUMBER() OVER(ORDER BY ROUND(AVG(freight_value) , 2) ) AS SL_NO
customer_state AS State,
ROUND(AVG(freight_value) , 2) AS Avg_freight_price
FROM
Target_Ecommerce.order_items o1
JOIN
Target_Ecommerce.orders o2 ON o1.order_id = o2.order_id
JOIN
Target_Ecommerce.customers c ON o2.customer_id = c.customer_id
GROUP BY
State
ORDER BY
Avg_freight_price
LIMIT 5
)

SELECT
*
FROM
Highest h
JOIN
Lowest l ON l.SL_NO = h.SL_NO
;
```

# Q11

Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

```
WITH
test1 AS
(
SELECT
  customer_state AS State,
  CEILING(AVG(
    DATE_DIFF(
      order_delivered_customer_date ,
      order_purchase_timestamp ,
      DAY
    ))) AS Avg_Delivery_Time,
  CEILING(AVG(
    DATE_DIFF(
      order_estimated_delivery_date ,
      order_purchase_timestamp ,
      DAY
    ))) AS Avg_Estimated_Delivery_Time
FROM
  Target_Ecommerce.orders o1
JOIN
  Target_Ecommerce.customers c ON o1.customer_id = c.customer_id
GROUP BY
  State
ORDER BY
  State
)

SELECT
  *,
  CEILING((Avg_Estimated_Delivery_Time - Avg_Delivery_Time)) AS Diff_Estimated_Delivery_Time
FROM
  test1
ORDER BY
  Diff_Estimated_Delivery_Time DESC
LIMIT 5;
```

| Row | State | Avg_Delivery_Time | Avg_Estimated_Delivery_Time | Diff_Estimated_Delivery_Time |
|-----|-------|-------------------|-----------------------------|------------------------------|
| 1   | AC    | 21.0              | 41.0                        | 20.0                         |
| 2   | RO    | 19.0              | 39.0                        | 20.0                         |
| 3   | AM    | 26.0              | 45.0                        | 19.0                         |
| 4   | AP    | 27.0              | 46.0                        | 19.0                         |
| 5   | RR    | 29.0              | 47.0                        | 18.0                         |

Insights:

- The difference in estimated delivery time is 20 days. This shows we are asking the customers for more time and giving the products before the asked time.

Recommendations:

- To reduce the difference we can discuss the time needed to deliver the product with the local delivery partner in the state and provide the estimation according.
- Sometimes customer might cancel the plan of ordering the items considering the duration to get the product.

# Q12

Find the month-on-month no. of orders placed using different payment types.

```
SELECT
    EXTRACT (Month FROM order_purchase_timestamp) AS Month,
    payment_type AS Mode_Of_Payment,
    COUNT(DISTINCT o.order_id) AS No_of_orders
FROM
    Target_Ecommerce.orders o
LEFT JOIN
    Target_Ecommerce.payments p ON p.order_id = o.order_id
GROUP BY
    Month, Mode_Of_Payment
ORDER BY
    Month, Mode_Of_Payment;
```

| JOB INFORMATION |       | RESULTS         | CHART        | JSON |
|-----------------|-------|-----------------|--------------|------|
| Row             | Month | Mode_Of_Payment | No_of_orders |      |
| 1               | 1     | UPI             | 1715         |      |
| 2               | 1     | credit_card     | 6093         |      |
| 3               | 1     | debit_card      | 118          |      |
| 4               | 1     | voucher         | 337          |      |
| 5               | 2     | UPI             | 1723         |      |
| 6               | 2     | credit_card     | 6582         |      |
| 7               | 2     | debit_card      | 82           |      |
| 8               | 2     | voucher         | 288          |      |
| 9               | 3     | UPI             | 1942         |      |
| 10              | 3     | credit_card     | 7682         |      |
| 11              | 3     | debit_card      | 109          |      |
| 12              | 3     | voucher         | 395          |      |
| 13              | 4     | UPI             | 1783         |      |

Insights:

- We can the majority of the payments were made using credit cards.
- Number of UPI payments are less than 30% for the credit card payments.

Recommendations:

- Get in touch with banks and establish a tie-ups and provide EMI offers to the customers.
- Give day wise offers on bank credit cards, so we can build a wider association with different banks.